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MILITARY SPECIFICATION

ICE CREAM FREEZERS, POWER OPERATED, MECHANICALLY REFRIGERATED: ICE CREAM CABINETS, MECHANICALLY REFRIGERATED: AND ICE CREAM PLANTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric-motor-driven, mechanically refrigerated, ice cream freezers; mechanically refrigerated, ice cream hardening cabinets; and freezers and hardening cabinets combined as complete, self-contained, mechanically refrigerated, ice cream plants.

1.2 Classification. Freezers, cabinets, and plants shall be of the following types and sizes, as specified (see 6.2.1).

Type I - Freezers.

Size 10 - 10-quart batch size.
Size 20 - 20-quart batch size.

Type II - Hardening cabinets.

Size 35 - 35-gallon storage capacity.
Size 60 - 60-gallon storage capacity.

Type III - Plants.

Size 10-35 - Type I, size 10 and type II, size 35.
Size 20-60 - Type I, size 20 and type II, size 60.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4110

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SPECIFICATIONS

FEDERAL

QQ-S-775 - Steel, Sheets, Carbon, Zinc-Coated (Galvanized) by the Hot Dip Process.

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-R-12323 - Refrigerator and Related Equipment, Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AIR-CONDITIONING AND REFRIGERATION INSTITUTE (ARI)

520 - Standard for Positive Displacement Refrigerant Compressor and Condensing Units.

(Application for copies should be addressed to the Air-Conditioning and Refrigeration Institute, 1815 North Fort Myer Drive, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASHRAE 34 - Number Designation of Refrigerants.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.
A176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ASSOCIATION OF MILK, FOOD, AND ENVIRONMENTAL SANITARIANS, INC. (IAMFES)

No. 19-03 - 3-A Sanitary Standards for Batch and Continuous Freezers for Ice Cream, Ices and Similarly-Frozen Dairy Foods.

(Application for copies should be addressed to the International Association of Milk, Food, and Environmental Sanitarians, Inc., Box 701, Ames, IA 50010.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 1 - Soda Fountain-Luncheonette Equipment and Appurtenances. Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, 3475 Plymouth, Ann Arbor, MI 48106.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 62 - Flexible Cord and Fixture Wire.
UL 471 - Commercial Refrigerators.
UL 498 - Electrical Attachment Plugs and Receptacles.
UL 621 - Ice Cream Freezers and Soda Fountain Units.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The ice cream freezers shall be of the batch type designed for operating cycles which include admitting one batch of the product to the freezing cylinder, partially freezing and incorporating air into the product to provide the overrun specified herein, adding flavoring materials and nuts or fruit, and discharging the product to containers for hardening. The hardening cabinets shall be of the size specified and shall be designed primarily for the hardening and storage of ice cream. When either freezers or hardening cabinets are procured as individual items, each item shall include a complete, self-contained refrigeration system. When freezers and cabinets are procured in combination as ice cream plants (type III), the plant shall include a single, self-contained condensing unit to service both the freezer and the cabinet.

* 3.2 First article. When specified (see 6.2.1), the contractor shall furnish one unit of each type and size being procured for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standards compliance. Equipment furnished under this specification shall meet the requirements of the following sanitary and safety standards, as applicable:

a. Freezers - IAMFES 3-A Sanitary Standards, No. 19-03 and UL 621.
b. Cabinets - NSF No. 1 and for cabinets of the air-cooled type employing hermetic compressor motors, UL 471.

* 3.3.1 Certification. Prior to approval of the first article, if one is submitted, or prior to the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the equipment meets the requirements of the applicable standards specified in 3.3. Acceptable evidence of meeting these standards will be as follows:

a. 3-A Sanitary Standards - The official 3-A symbol on the equipment and a listing by the 3-A Sanitary Standards Symbol Administrative Council.
b. NSF No. 1 - Listing of the model being furnished in the current edition of the NSF Listing of Food Service Equipment and authorized display of the NSF seal on the finished item.
c. UL 471 and UL 621 - A UL approved listing and listing mark.

* 3.3.2 Alternate certification. In lieu of the certification required under 3.3.1, acceptable evidence of meeting the requirements of the standards cited will be a certified test report from a recognized independent testing laboratory, selected by the manufacturer and approved by the contracting officer, or in the case of 3-A and NSF certification, approved by the cognizant medical authority acting in an advisory capacity to the contracting officer, indicating a freezer or cabinet identical to the items to be furnished under this specification has been inspected in accordance with the applicable standard and meets the requirements thereof. Alternate certification shall be submitted prior to approval of the first article or prior to commencing production unless freezers or cabinets will be furnished from the contractor's current inventory, in which case alternate certification shall be submitted prior to the first shipment.

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3.3.3 Applicability. Details for equipment construction, design, and performance covered by this specification shall be supplementary to the requirements of the standards cited in 3.3 and in compliance with and approval under these standards shall not relieve the contractor of the responsibility for meeting the requirements of this specification.

* 3.4 Standard commercial product. The freezers, cabinets, or plants shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the freezers, cabinets, or plants being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.5 Interchangeability of items. All equipment of the same classification furnished under any specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, and spare parts. No deviations will be acceptable without prior written approval of the contracting officer. All threaded parts shall conform to FED-STD-H28.

* 3.6 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.6.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fastenings shall be of the same material as the metals joined or shall be cathodic to the metals joined.

* 3.6.2 Galvanized sheet steel. When galvanized steel sheets are specified herein, they shall conform to the requirements for class d of QQ-S-775.

* 3.6.3 Corrosion-resistant steel. Corrosion-resistant steel shall be class 302, 303, or 304 of ASTM A167 or class 430 of ASTM A176 annealed or tempered as required to suit the intended application. Corrosion-resistant material for product contact surfaces shall be as specified in IAMFES 3-A Sanitary Standard, No. 19-03.

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* 3.6.4 Insulation. Insulation shall have a commercially rated thermal conductivity not exceeding 0.275 British thermal units per hour per square foot per inch of thickness per degree Fahrenheit (°F). Insulation shall be of such material and installed in such a manner that occurrences of gaps or open spaces between parts of the insulation will not exist; it shall remain in place and shall be effectively and permanently sealed from seepage and condensation. The thickness of insulation shall be sufficient to insure compliance with all performance requirements of 3.12.

* 3.6.5 Refrigerant. The refrigerant shall be dichlorodifluoromethane (R 12) or chloropentafluoroethane (R502). The refrigerants shall conform to applicable requirements of ANSI/ASHRAE 34.

3.7 Capacity. The minimum capacity of the equipment shall be as specified in 1.2. For freezers, the capacity shall be designated by the batch size. One batch shall consist of the volume in gallons of the product as discharged with 100 percent overrun from the freezer after one cycle is completed. For hardening cabinets, the minimum capacity shall be as specified in 1.2 and shall be based on the use of round 2-1/2 gallon containers.

* 3.8 Design. The equipment and accessories shall be designed to prevent conditions hazardous to personnel or deleterious to equipment, shall withstand the hard usage encountered in military service, and shall permit easy accessibility for maintenance and service in the field.

3.9 Construction.

3.9.1 Type I. The type I freezer shall consist essentially of a cabinet enclosure, self-contained condensing unit, freezing cylinder, electric-motor-driven beater, automatic temperature controls, and necessary valves, piping and associated appurtenances. The cabinet shall form a rigid structure having the refrigeration system in the base and the freezing section in the top. The outer shell of the cabinet shall be fabricated of corrosion-resistant steel specified in 3.6.3. The freezing cylinder shall be horizontally mounted and shall be equipped with one or more scraper blades. The beater motors shall have minimum continuous horsepower (hp) ratings of 1 hp and 1.5 hp for sizes 10 and 20 respectively. The freezer shall be furnished with a gravity-feed filler tube or spout suitable for charging the cylinder with mix, a removable or hinged cylinder head, a quick-operating dispensing valve, and a container shelf located under the dispensing valve. The freezer shall otherwise be constructed in accordance with IAMFES 3-A Sanitary Standard, No. 19-03.

3.9.2 Type II. Type II hardening cabinets shall consist essentially of a refrigerated, ice cream storage compartment or compartments with inner liner and outer shell; a self-contained, automatically-controlled condensing unit with evaporator sections sized and located as required to meet the performance requirements of 3.12; panels for access to the condensing unit, lids, and insulation conforming to 3.6.4. The cabinet shall be constructed to support a full load of ice cream without deformation of the inner bottom liner or other supporting panels or structural members. The outer shell shall be as follows:

a. Top - Corrosion-resistant steel; one-piece construction unless full-depth lids form the top of the cabinet.
b. Front and ends - Corrosion-resistant steel or baked porcelain enamel.

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c. Rear and bottom - Corrosion-resistant steel or baked porcelain enamel, or galvanized or galvannealed sheet steel.

The inner liner of the storage compartment(s) shall be corrosion-resistant steel or galvanized or galvannealed sheet steel at the option of the contractor. Lids shall be fabricated of corrosion-resistant steel, shall be insulated, and shall be of the double section type with integral dividing hinge or shall be of the single section type, hinged to the cabinet at the rear, and providing complete access from the front to the rear of the cabinet compartment. Legs, if used, shall conform to NSF No. 1. The thickness of materials used for fabrication of the cabinet, including the outer shell, inner liner, and lids shall be the thickness used on the models approved under the standards cited in 3.3.

3.9.3 Type III. Type III plants shall consist of a freezer and a hardening cabinet. Except as otherwise specified herein, the freezer and cabinet shall meet the requirements of 3.9.1 and 3.9.2, respectively. Each plant shall include one condensing unit to service both the freezer and the cabinet. The condensing unit may be mounted either in the base of the freezer as specified in 3.9.1 or in the hardening cabinet. If the condensing unit is located in the hardening cabinet, the freezer shall be furnished without a base section and shall be mounted on the top of the cabinet over the condensing unit compartment. When the freezer is cabinet mounted, the container shelf specified in 3.9.1 shall be secured to the front of the cabinet under the dispensing valve. Appropriate refrigeration controls shall be provided to automatically maintain the operating temperatures specified herein.

3.9.3.1 Skid mounting. When specified (see 6.2.1), type III plants shall be skid mounted. The skid shall consist essentially of two structural aluminum I-beam runners mounted longitudinally under the front and rear edges of the hardening cabinet. The runners shall be equal in length to the cabinet except that, when the freezer is separate rather than cabinet mounted, the runners shall be extended sufficiently to accommodate the freezer. The I-beam runners shall be a minimum of 5 inches in depth by 3 inches in width and shall be upturned or beveled at each end to facilitate skidding. Towing holes not less than 2 inches in diameter shall be provided at both ends of each runner. The skids shall be welded or bolted to framing members of the cabinet with suitable aluminum cross bracing or mounting plates provided as required. The skid base shall have sufficient strength and rigidity to permit skidding of the complete plant into place on level ground without damage or deformation to the skid base or to the components which the skid base supports.

* 3.10 Refrigeration system. The refrigeration system shall consist essentially of an electrically-operated compressor, a condenser, evaporator sections with capillary tube or thermal expansion valve, automatic thermostatic controls, dehydrator (drier), liquid indicator, and all necessary valves, fittings, and piping. Unless water-cooled condensing units are specified (see 6.2.1), the condensers shall be air cooled. The condensing units shall be constructed and rated in accordance with ARI 520. The refrigeration system shall have sufficient hp and refrigeration capacity to meet the requirements of 3.12 without exceeding the established hp and thermal-capacity ratings or the UL approved electrical rating. Unless otherwise specified (see 6.2.1), condensing units rated at 3/4 hp or less shall be designed for

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operation on a nominal 120 volt (V), 60 Hertz (Hz), 3 phase supply. The single condensing unit for type III plants shall have sufficient capacity to permit simultaneous operation of both the freezer and the hardening cabinet. Refrigerant systems shall be completely dehydrated and fully charged prior to shipment.

* 3.11 Electrical wiring and components. All wire, cord, and electrical components shall be installed in accordance with NFPA No. 70. Electric cords shall conform to UL approved, 2 pole, 3 wire, grounding type plug conforming to UL 498 having dead front construction. A wiring diagram shall be furnished with each unit.

3.12 Performance. Freezers and hardening cabinets, furnished separately or in combination, as a complete plant shall meet the performance requirements herein under the specified ambient temperatures.

3.12.1 Freezers. Freezers shall be capable of producing, in a period not exceeding 10 minutes, one batch of ice cream from a vanilla mix having a temperature, when introduced into the freezer, of 38°F and a temperature when discharged, not exceeding 24.5°F. The vanilla mix shall be a standard commercial mix consisting of 10 percent butterfat with a minimum total solids of 20 percent. During the freezing process, sufficient air shall be introduced by the beater into the product to provide overrun in the range of 90 to 100 percent by volume (50 percent mix and 40 to 50 percent air). These performance requirements shall be met in an ambient temperature of 100°F, after the freezing cylinder has been stabilized at the normal operating temperature.

3.12.2 Hardening cabinets. Hardening cabinets shall meet the following performance requirements:

a. In an ambient temperature of 100°F, the refrigeration system shall be capable of lowering the temperature of the empty cabinet from 100°F, to -10°F in not more than 6 hours.
b. When the cabinet temperature has been stabilized at -10°F, the maximum operating time in an ambient temperature of 100°F under no load conditions shall be 80 percent over any 6-hour period.
c. After the cabinet temperature has been stabilized at -10°F in an ambient temperature of 100°F, the cabinet shall be capable of reducing the temperature of a full load of ice cream from 24.5°F to 0°F or below in not more than 16 hours.
d. Hardening cabinets shall be capable of operating without exceeding the rated capacity of the condensing unit in ambient temperatures up to 110°F, and under ambient conditions of 65°F to 75°F, shall be capable of maintaining a cabinet temperature of -20°F.

* 3.13 Identification marking. Identification shall be permanently and legibly marked directly on the cabinet or on a corrosion-resisting metal plate securely attached to the cabinet at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

* 3.14 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

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a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

* 3.15 Finish. The exterior cabinet surfaces and the interior compartment surfaces shall be finished in accordance with the manufacturer's standard practice. Concealed steel supporting shapes and plates shall be finished as specified in NSF No. 1.

* 3.16 Commercial publications. Publications furnished with each cabinet shall be the manufacturer's standard commercial publications as specified (see 6.2.1 and 6.3).

3.17 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall be designed and equipped to meet the electromagnetic interference control requirements and test limits for class III equipment as specified MIL-STD-461 when tested in accordance with 4.5.4, C3, group 3.

* 3.18 Workmanship. The finished equipment shall be clean inside and outside. All exposed sharp edges, burrs, loose solder, weld splatter, and trimming marks shall be removed. The quality of workmanship shall otherwise be of the level applied to freezers and cabinets sold on the commercial market and as established by UL, 3-A, and NSF for the approval of freezers and cabinets.

* 3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

* 3.18.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

* 3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one unit of each type and size when a first article is required (see 3.2 and 6.2.1). This inspection shall include the examination of 4.4, the tests of 4.5.1, 4.5.2, and 4.5.3 and when electromagnetic interference control is specified (see 6.2.1), the test 4.5.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the test of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling for examination. All freezers, hardening cabinets, and plants of the same type and size shall be in accordance with MIL-STD-105. All freezers, hardening cabinets, and plants offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level shall be 2.5 for major defects and 4.0 for minor defects in terms of defects per hundred units.

4.4 Examination. Each item selected shall be examined for the applicable requirements of this specification and for defects listed in table I.

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TABLE I. Classification of defects.

Classification	Defects	Requirement paragraph
Major:		
101	Units not type and size specified.	1.2 and 3.1
102	Items not interchangeable.	3.5
103	Material not as specified: defective materials affecting usability of the unit.	3.6
104	Refrigerant not of the type specified.	3.6.5
105	Minimum capacities not as specified.	3.7
106	Design not as specified.	3.8
107	Type freezer not as specified.	3.9.1, 3.9.2 and 3.9.3
108	Construction not in accordance with 3-A Sanitary Standards, or NSF No. 1, as applicable.	3.9.1 and 3.9.2
109	Appropriate automatic refrigeration controls not provided.	3.9.3
110	Refrigeration system not as specified.	3.10
111	Electric wire, cord, and electrical components not as specified; wiring diagrams not furnished with each unit.	3.11
112	Performance of units not as specified.	3.12
113	Concealed steel supporting shapes and plates not finished as specified in NSF No. 1.	3.15
114	Workmanship not as specified.	3.18
Minor:		
201	Marking not as specified.	3.13
202	Commercial publications not furnished with each unit.	3.16

4.4.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.3.1. The Government reserves the right to examine and test all freezers, cabinets, and plants to determine the validity of the certification.

4.5 Tests.

4.5.1 Functional test. The first article and each freezer and cabinet shall be operated for a length of time sufficient to insure, by visual inspection of the liquid indicator, that the refrigeration system is dehydrated, that switches and controls are correctly adjusted and function properly, that the refrigeration system is leaktight, and that mechanical components, such as the freezer beater, operate properly.

* 4.5.2 Freezer test. The first article freezer unit shall be operated for at least 1 hour using the vanilla mix specified in 3.12.1. The unit shall be checked for capacity and the final product checked for temperature and overrun.

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* 4.5.3 Cabinet test. The first article hardening cabinet shall be operated for a period of time sufficient to verify compliance with the performance requirements of 3.12.2 a., b., and c. The tests shall include the no-load pull down test, compliance with 3.12.2 b. shall be verified. The tests shall be conducted in the ambient temperatures specified. For the full load tests, ice cream in 2-1/2 gallon containers, or a liquid having the same specific heat may be used. When a simulated load is used, the total weight of the load shall be equivalent to the weight of a full load of ice cream having 100 percent overrun. After the full load test is completed, compliance with the temperature requirement of 3.12.2 d. shall be verified in an ambient temperature between 65°F and 75°F.

4.5.4 Electromagnetic interference control tests. When electromagnetic interference control is specified (see 6.2.1), and the condensing units furnished are capable of inducing conducted or radiated interference, the first article sample shall be tested in accordance with the methods of MIL-STD-462 to verify compliance with the test limits of MIL-STD-461 as specified (see 3.17).

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-R-12323. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2.1), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2.1).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-R-12323 with the level of preservation and the level of packing as specified (see 6.2.1).

6. NOTES

6.1 Intended use.

6.1.1 Freezers. Freezers covered by this specification are intended for the production of ice cream, ices, and similarly frozen dairy foods which are subsequently hardened in the type II cabinets and are ultimately served as "hard" ice cream. The freezers are not designed for the production of "soft" ice cream served directly from the freezer.

6.1.2 Hardening cabinets. Hardening cabinets covered by this specification are designed for hardening and storing ice cream and are not intended for dispensing purposes. Cabinets suitable for dispensing are covered by MIL-I-20565. Use of ice cream cups, such as those conforming to MIL-C-15820, or use of 1-quart containers for sliced portions (six to eight slices per container) minimizes the need for separate dispensing facilities.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2).

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c. When a first article sample is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When type III plants shall be skid mounted (see 3.9.3.1).
e. When water-cooled in lieu of air-cooled condensing units are required (see 3.10).
f. When power supply for units of 3/4 hp or less will be other than 120 V, 60 Hz, 3 phase (see 3.10).
g. When fungus treatment is required (see 3.14).
h. Commercial publications required (see 3.16).
i. When electromagnetic interference control is required (see 3.17 and 4.5.4).
j. When packaging inspection is required (see 4.6).
k. Level of preservation and level of packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.	Option
3.16	Publications, Commerical	DI-M-24010A	

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item consisting of one complete unit of each type and size or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

* Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

Project No. 4110-0372

Review Activity:

DLA - GS

User Activities:

Navy - MS, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

4110-0372

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-16042E	MIL-I-16042E - ICE CREAM FREEZERS, POWER OPERATED, MECHANICALLY REFRIGERATED & ICE CREAM CABINETS,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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